Thiranex Internship

Task 3: Predictive Analytics Using Historical Data

Objective: Build a predictive model using historical data to estimate future revenue trends, preprocess the dataset, evaluate model performance, and visualize predictions.

1. Dataset Overview

The demonstration dataset contains 69 monthly records from Jan 2021 to Sep 2026. Fields include date, marketing spend, units sold, and revenue.

2. Methodology

The date field was transformed into a time index. Month number and sine/cosine features represent seasonality. The first 80% of records were used for training and the final 20% for testing, preserving chronological order. A Linear Regression model was trained and evaluated on the held-out period.

3. Model Evaluation

Model: Linear Regression
Test MAE: ■742.80
Test RMSE: ■1,009.18
Test R^2: 0.802
Forecast horizon: 12 months

MAE is average absolute prediction error. RMSE gives larger errors more weight. R^2 indicates how much variation the model explains on the test period. Scores should be compared with simple forecasting baselines before real use.

4. Visualizations

The dashboard compares historical revenue with held-out predictions and displays the next 12 months of forecast values.

5. Recommendations

- Compare results with a last-month and seasonal-naïve baseline.
- Update the model when new data becomes available.
- Monitor forecast errors and investigate large deviations.
- Validate results with real historical data before business decisions.

6. Deliverables

Interactive HTML dashboard, historical CSV, test prediction CSV, 12-month forecast CSV, Excel workbook, Python script, requirements file, preview image, and this report.

Important Data Note

No real historical dataset was included with the task screenshot. This project uses synthetic data generated for demonstration. Replace it with a dataset provided by Thiranex, if available, before presenting these as actual business forecasts.